

High-Speed Generative Diffusion and DDIM Deterministic Sampling for Risk-Aware Handover in 5G Networks

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High-Speed Generative Diffusion and DDIM Deterministic Sampling for Risk-Aware Handover in 5G Networks

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of the requirements for the degree*

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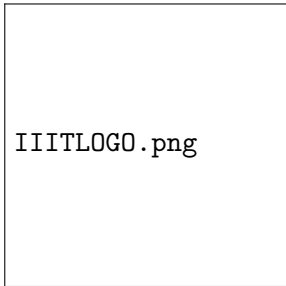
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Abstract

The transition of cellular networks from 5G to next-generation 6G systems demands a fundamental paradigm shift in mobility management. Traditional reactive handover (HO) procedures, which rely on signal strength comparisons and hysteresis parameters sustained over a Time-To-Trigger (TTT) window (e.g., 3GPP Event A3), are increasingly insufficient for millimeter-wave (mmWave) and terahertz (THz) deployments. In these high-frequency bands, obstacle blockages can cause instantaneous Reference Signal Received Power (RSRP) drops of 20–30 dB within milliseconds, leading to severe Radio Link Failures (RLFs) or frequent ping-pong triggers.

This thesis develops and evaluates conditional generative diffusion-based probabilistic forecasting models for proactive handover prediction. Instead of predicting a single deterministic future, our framework generates an ensemble of $N = 50$ probabilistic future RSRP trajectories, enabling the network control plane to perform risk-aware decisions by evaluating the exact probability of connection failure, defined as $P(\text{RSRP} < -85 \text{ dBm})$.

We implement and evaluate two parallel diffusion configurations on a commercial LTE drive test dataset: (i) a parallel Denoising Diffusion Probabilistic Model (DDPM), and (ii) a fast deterministic Denoising Diffusion Implicit Model (DDIM).

Our experimental results demonstrate that for a short prediction horizon (10 ms), deterministic DDIM sampling (50 steps) achieves a **1.97 $\times$ speedup** over standard DDPM (100 steps) during inference (0.1887 seconds vs. 0.3712 seconds per window) with negligible degradation in forecasting quality (+0.11 dB in MAE). Crucially, DDIM achieves the lowest False Negative Rate of **14.29%**, demonstrating superior signaling safety. Finally, we propose a Software Defined Networking (SDN) deployment architecture, hosting the forecasting engine as a centralized application on top of the SDN controller to coordinate proactive network-wide handovers and reconfigure forwarding paths before RLF events occur, meeting the latency constraints of high-speed mobile nodes.

Keywords: *Handover prediction, mmWave, Denoising Diffusion Probabilistic Models (DDPM), Denoising Diffusion Implicit Models (DDIM), Continuous Ranked Probability Score (CRPS), Software Defined Networking (SDN), SDN Controller.*

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List of Abbreviations

3GPP	: 3rd Generation Partnership Project
AI	: Artificial Intelligence
AMF	: Access and Mobility Management Function
CDF	: Cumulative Distribution Function
CRPS	: Continuous Ranked Probability Score
DDIM	: Denoising Diffusion Implicit Model
DDPM	: Denoising Diffusion Probabilistic Model
DL	: Deep Learning
ECE	: Expected Calibration Error
eNB	: Evolved Node B (4G LTE Base Station)
FN	: False Negative
FNR	: False Negative Rate
FP	: False Positive
gNB	: Next Generation Node B (5G Base Station)
GRU	: Gated Recurrent Unit
HO	: Handover
HOSR	: Handover Success Rate
KPI	: Key Performance Indicator
LTE	: Long Term Evolution
LSTM	: Long Short-Term Memory
MAE	: Mean Absolute Error
ML	: Machine Learning
mmWave	: Millimeter Wave
MSE	: Mean Squared Error
MTS	: Minimum Time of Stay
NR	: New Radio
PPR	: Ping-Pong Rate
QoS	: Quality of Service
RACH	: Random Access Channel
RLF	: Radio Link Failure
RMSE	: Root Mean Square Error
RNN	: Recurrent Neural Network
RSRP	: Reference Signal Received Power
SDN	: Software Defined Networking
THz	: Terahertz
TTT	: Time-To-Trigger
UE	: User Equipment
UPF	: User Plane Function
URLLC	: Ultra-Reliable Low-Latency Communications

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Introduction

Preface

In this chapter, we present the introduction to proactive handover prediction in 5G and beyond networks. We discuss the challenges of mmWave band signal propagation, the limitations of standard 3GPP Event A3 reactive triggers, and motivate the need for generative diffusion-based probabilistic forecasting models to enable risk-aware network decision making. We outline the key contributions of this thesis and the organization of the chapters.

1.1 Introduction

Mobility management is one of the core pillars of cellular networks, ensuring that User Equipment (UE) can maintain continuous connectivity while moving across different cells. As cellular networks transition from 5G to next-generation 6G systems, the deployment of high-frequency bands, such as millimeter-wave (mmWave) and terahertz (THz) spectrums, has become essential to support high data rates, massive network capacity, and ultra-reliable low-latency communications (URLLC) [1]. However, these high-frequency bands present severe signal propagation challenges that require a fundamental rethink of standard mobility management protocols.

1.1.1 Physical Limitations of high-frequency Bands

Millimeter-wave signals (operating in Frequency Range 2 (FR2), typically between 24 GHz and 52 GHz) and THz signals operate at very short wavelengths. According to Friis free-space transmission equation, the path loss PL scales quadratically with the carrier frequency f :

$$PL(d, f) = 20 \log_{10}(d) + 20 \log_{10}(f) + 20 \log_{10} \left(\frac{4\pi}{c} \right) \quad (1.1)$$

where d is the distance between transmitter and receiver, and c is the speed of light. Consequently, high-frequency signals suffer from severe isotropic path loss, atmospheric absorption, and rain attenuation, restricting coverage areas to small cells (typically 100–200 meters in radius).

Furthermore, high-frequency bands exhibit extremely poor penetration and diffraction characteristics. Common urban building materials cause substantial signal degradation:

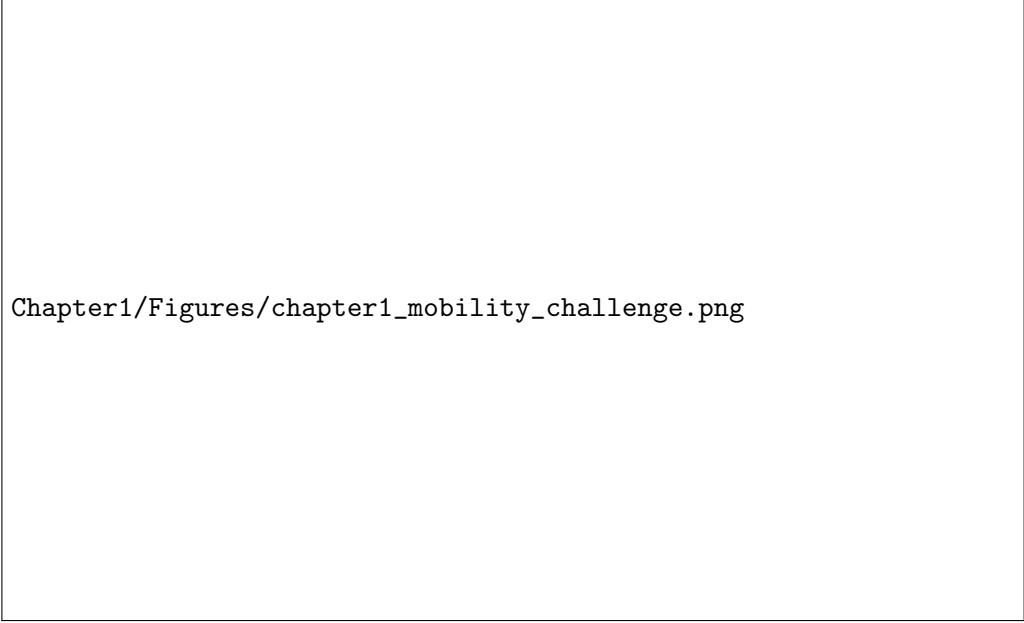
- **Concrete Walls:** Introduce a signal attenuation of 35–50 dB.
- **Tinted Glass:** Introduces a loss of 20–30 dB.
- **Drywall and Wood:** Cause a loss of 5–10 dB.

When a mobile user moves behind an obstacle (e.g., turning a street corner or passing behind a large vehicle), the Line-of-Sight (LoS) path is instantaneously severed. This results in a sudden, sharp signal drop known as shadow fading or blockage, where the Reference Signal Received Power (RSRP) can drop by 20–30 dB within a few milliseconds (as illustrated in Figure 1.1).

1.1.2 Limitations of Standard Reactive Handover

In traditional LTE and sub-6 GHz 5G networks, the handover process is reactive and threshold-based. Under the 3GPP standards, measurement reports are evaluated against specific event conditions, such as Event A3. This event triggers when a neighboring target cell's RSRP exceeds the serving cell's RSRP by a hysteresis margin. To prevent transient channel noise from triggering premature handovers, this condition must remain satisfied for a continuous duration known as the Time-To-Trigger (TTT).

In high-frequency environments, the rapid signal degradation caused by blockages occurs on a millisecond timescale, which is much faster than the standard TTT window (typically configured between 80 ms and 5120 ms). By the time the TTT timer expires and the UE transmits a Measurement Report, the signal quality of the serving cell has often dropped below the receiver sensitivity threshold, leading to a **Radio Link Failure (RLF)** and service disruption. Conversely, setting overly aggressive margins or lowering the TTT leads to frequent **ping-pong handovers**, where the UE repeatedly switches back and forth between adjacent cells, wasting control plane signaling bandwidth and degrading the overall Quality of Service (QoS) [2, 3].



Chapter1/Figures/chapter1_mobility_challenge.png

Figure 1.1: 5G mmWave mobility challenge showing severe shadow fading and a 30 dB signal drop due to building blockage along the user trajectory.

1.1.3 Proactive Handover and Generative Diffusion Models

Proactive handover prediction reframes mobility management as a time-series forecasting problem. By predicting the future RSRP trajectories of serving and neighboring cells before signal degradation occurs, the network can proactively prepare target base stations and execute handovers in advance [4, 5, 6]. This thesis builds directly upon the foundational frameworks of two key literature baselines: the two-stage machine learning prediction model proposed by Lima et al. [7] ("Deep Learning-Based Handover Prediction for 5G and Beyond Networks"), and the comprehensive time-series evaluation model proposed by Park et al. [8] ("A Comprehensive Evaluation of Time-Series Forecasting Methods for Proactive Handover").

This thesis focuses on developing and evaluating conditional generative diffusion-based probabilistic forecasting models to generate multiple probabilistic future RSRP scenarios, enabling risk-aware handover decisions that are crucial for safety-critical and low-latency network orchestrations. By modeling the future as a distribution, the network can make optimal mobility decisions that balance the risk of RLFs against the cost of premature handovers. We implement these models within a centralized Software Defined Networking (SDN) architecture, hosting the forecasting engine as an application on top of the SDN controller to coordinate proactive network-wide handovers and reconfigure forwarding paths before RLF events occur, meeting the latency constraints of high-speed mobile nodes [9].

1.2 Motivation

Traditional deep learning approaches to handover prediction rely heavily on deterministic point forecasting models, such as Long Short-Term Memory (LSTM) networks, Gated Recurrent Units (GRUs), or linear time-series models (e.g., DLinear/NLinear). While these models can predict a single future RSRP value or sequence, they lack native uncertainty quantification. In highly dynamic wireless channels, a single deterministic prediction is insufficient for risk-critical mobility orchestration. For example, if a model predicts a future RSRP of -90 dBm, the network cannot determine the probability that the signal will drop below the critical failure threshold (e.g., -95 dBm) or recover.

1.2.1 Aleatoric Uncertainty in Wireless Channels

Wireless signal propagation is inherently stochastic due to multipath reflection, scattering, refraction, and dynamic physical blockages. This uncertainty is primarily **aleatoric** (inherent physical randomness) rather than epistemic (model uncertainty).

A deterministic network trained with Mean Squared Error (MSE) loss acts as an expected-value estimator, predicting the mathematical mean of all possible future paths:

$$\hat{Y} = \mathbb{E}[Y|X_{\text{hist}}] \quad (1.2)$$

Here, $\hat{Y} \in \mathbb{R}^{W_{\text{out}} \times 1}$ represents the predicted future RSRP sequence, $\mathbb{E}[\cdot|\cdot]$ denotes the conditional mathematical expectation operator, $Y \in \mathbb{R}^{W_{\text{out}} \times 1}$ represents the true actual future RSRP trajectory, and $X_{\text{hist}} \in \mathbb{R}^{W_{\text{in}} \times 1}$ represents the historical sequence of observed RSRP measurements.

Because the mean of a multi-modal distribution (e.g., a wireless channel where the signal either continues stably or drops sharply by 30 dB due to building blockages) is a smoothed, intermediate path, deterministic models consistently fail to predict the sudden, sharp signal drops that lead to RLFs. They output a smoothed average signal level that appears stable, blinding the network controller to the catastrophic tail risk of connection loss.

1.2.2 Expected-Cost Minimization in Network Orchestration

Probabilistic forecasting models, such as Denoising Diffusion Probabilistic Models (DDPMs), address this limitation by generating a distribution of potential future trajectories rather than a single average path. This distributional output allows the network to calculate the exact probability of failure, defined as $P(\text{RSRP} < \gamma)$, where the outage threshold γ is set to -85 dBm (with a typical operational range between -85 dBm and -95 dBm in standard commercial user equipment, below which radio links are highly prone to disconnection).

With this probabilistic information, an SDN network controller can make optimal risk-based routing decisions. In cellular networking, the cost of a Radio Link Failure (C_{RLF}) in terms of signaling re-establishment and packet loss is orders of magnitude higher than the cost of proactive resource preparation at a target cell (C_{prep}):

$$C_{\text{RLF}} \gg C_{\text{prep}} \quad (1.3)$$

A risk-aware controller evaluates the expected cost of routing actions:

$$\mathbb{E}[\text{Cost}] = P_{\text{failure}} \cdot C_{\text{RLF}} + (1 - P_{\text{failure}}) \cdot C_{\text{prep}} \quad (1.4)$$

By setting decision thresholds that minimize this expected cost, the controller can initiate resource preparation even when the probability of failure is low (e.g., $P_{\text{failure}} = 0.15$), safeguarding connection continuity. Developing and evaluating these generative probabilistic models for proactive 5G handover prediction forms the core motivation of this research.

1.3 Thesis Contributions

The primary contributions of this thesis are summarized as follows:

1. **Conditional DDPM for RSRP Forecasting:** We propose a conditional Denoising Diffusion Probabilistic Model (DDPM) that takes historical RSRP measurements and generates multiple probabilistic future scenarios. This provides native uncertainty quantification and enables risk-aware handover decisions without requiring separate classification stages. The model is conditioned on a recurrent history representation, allowing it to capture macro-mobility trends.

2. **DDIM Inference Optimization:** We apply Denoising Diffusion Implicit Models (DDIM) sampling to the trained DDPM, reducing the number of inference denoising steps from 100 to 50. This achieves a $\sim 2\times$ inference speedup with negligible accuracy degradation, making it suitable for real-time control-plane constraints. We demonstrate that deterministic sampling ($\eta = 0$) stabilizes the predicted trajectory distributions and reduces the rate of missed triggers.
3. **SDN Deployment Plan and Latency Feasibility Study:** We formulate a centralized Software Defined Networking (SDN) deployment plan to integrate the fast DDIM inference engine on top of the SDN controller. We analyze the components, data flow, and OpenFlow mechanisms, and evaluate the physical latency feasibility of the control loop against physical velocity deadlines, establishing a Pareto-optimal model for network deployment.

1.4 Thesis Outline

The remainder of this thesis is organized as follows:

- **Chapter 2: Background & Literature Review** provides the theoretical foundations for this work. We review 3GPP LTE and 5G New Radio handover protocols, signaling flows, and the mechanics of Event A3. We then analyze deep learning baselines, detailing the regression-classification pipeline from Lima et al. [7] and time-series benchmarks from Park et al. [8] Finally, we introduce the mathematical theory of denoising diffusion models (DDPM and DDIM).
- **Chapter 3: Methodology** details the formal implementation of our forecasting models. We formalize proactive handover as a conditional probabilistic forecasting problem. We then present the network architecture of the conditional **RSRPDiffusion** model We detail the deterministic DDIM step-skipping schedule.
- **Chapter 4: Experimental Setup** describes the empirical environment used for validation. We detail the commercial LTE drive-test dataset collected in Belo Horizonte, Brazil, the data collection campaign, and preprocessing steps. We then define the quantitative evaluation metrics (MAE, CRPS with its accuracy/diversity decomposition, False Negative Rate, and Inference Latency).
- **Chapter 5: Results & Analysis** presents the comparative evaluation of the models. We analyze the performance of DDPM, DDIM and DDIM across a short prediction horizon (10 ms) and a long prediction horizon (50 ms). We analyze the error distributions, cumulative errors, probability calibration, and CRPS decompositions, and present the thesis-quality figures generated during the evaluation.
- **Chapter 6: Discussion** analyzes the deployment feasibility. We present a centralized Software Defined Networking (SDN) deployment architecture, describing the components, data flows, and OpenFlow northbound/southbound interactions. We calculate the total control loop latency budget and discuss the trade-offs between prediction speed and quality.
- **Chapter 7: Conclusion & Future Work** summarizes the key findings of this thesis and outlines future avenues of research, focusing on beam-level predictions in Frequency Range 2, distilled consistency models for sub-millisecond inference, and federated learning for domain adaptation.

Background & Literature Review

◇

Preface

In this chapter, we establish the theoretical foundations and review the related work for proactive handover prediction in 5G networks. We discuss the physical mechanics of signal propagation and cell-switching protocols under 3GPP standards, analyze previous deep learning-based forecasting and classification baselines, and review the mathematical theory of denoising diffusion probabilistic models.

◇

2.1 3GPP Handover Mechanics and Challenges

In mobile communication networks, handover is the mechanism by which a User Equipment (UE) maintains its active sessions while transitioning across different coverage areas. In 3GPP LTE and 5G New Radio (NR) networks, this process is governed by standardized measurement events where the UE periodically measures the Reference Signal Received Power (RSRP) of serving and neighboring base stations. The network leverages these measurements to ensure optimal radio link quality and to prevent service interruptions as the user moves.

2.1.1 3GPP Handover Measurement and Signaling Flow

The standard 3GPP handover execution involves a sequence of coordination phases between the UE, the serving base station (source gNB), and the candidate base station (target gNB), orchestrated under control plane protocols. The procedure is broadly divided into three main phases:

1. **Measurement Configuration:** The source gNB configures the UE's measurement parameters, including the reporting criteria, the carrier frequencies to monitor, and the specific event thresholds. The UE filters physical layer samples using a dual-stage filter (Layer 1 and Layer 2 filtering) to smooth out fast fading effects and estimate the macro-level RSRP.
2. **Handover Decision and Resource Preparation:** Once a reporting event condition is satisfied, the UE transmits a Measurement Report to the source gNB. The source gNB evaluates the report and decides whether to initiate a handover. If it decides to proceed, it sends a Handover Request to the target gNB over the Xn interface (in 5G) or X2 interface (in LTE). The target gNB performs admission control and allocates radio resources (such as Dedicated Random Access Preamble and Cell Radio Network Temporary Identifier (C-RNTI)). It then sends a Handover Request Acknowledge back to the source gNB.
3. **Handover Execution and Completion:** The source gNB transmits the Handover Command (contained in an `RRCReconfiguration` message) to the UE. Upon receiving this command, the UE detaches from the source gNB, synchronizes with the target gNB using the random access channel (RACH), and transmits an `RRCReconfigurationComplete` message. Finally, the target gNB notifies the User Plane Function (UPF) and Access and Mobility Management Function (AMF) to switch the user plane path, completing the handover.

2.1.2 Standard Event A3 Triggering Condition

Handover execution is primarily triggered by **Event A3**, which occurs when a neighbor cell's RSRP becomes offset-better than the serving cell's RSRP. Mathematically, the entering condition for Event A3 is defined as:

$$M_{\text{target}} + O_{\text{target}} - Hys > M_{\text{serving}} + O_{\text{serving}} + Off \quad (2.1)$$

where M_{target} and M_{serving} denote the measured RSRP of the candidate and serving cells in dBm, respectively. The parameter Off represents the event-specific offset, Hys is the hysteresis parameter (used to prevent ping-pong effects by requiring the target cell to be significantly better), and O_{target} and O_{serving} represent cell-specific and frequency-specific offsets.

Conversely, the leaving condition for Event A3, which determines when the event is cancelled if the signal fluctuates back, is defined as:

$$M_{\text{target}} + O_{\text{target}} + Hys < M_{\text{serving}} + O_{\text{serving}} + Off \quad (2.2)$$

To prevent transient noise and short-term fading from triggering unnecessary handovers, the entering inequality condition must be sustained continuously for a configured duration known as the **Time-To-Trigger (TTT)** (typically ranging from 40 ms to 5120 ms). Once the TTT timer expires, the UE sends the Measurement Report to the serving base station, which then coordinates resource preparation.

2.1.3 Challenges in High-Frequency (mmWave/THz) Bands

While this reactive threshold-based mechanism is highly robust in conventional sub-6 GHz frequencies, it is increasingly inadequate for millimeter-wave (mmWave) and terahertz (THz) deployments [10, 3, 11]. In mmWave bands (such as 28 GHz or 39 GHz), wireless propagation is highly directional and suffers from severe path loss and penetration loss. Large obstacles, such as vehicles, buildings, or even pedestrians blocking the Line-of-Sight (LoS) path, can cause the RSRP to drop abruptly by 20–30 dB within milliseconds.

Under high mobility scenarios (e.g., UEs traveling at 50–60 km/h in urban streets), the duration of this signal drop is much shorter than the standard TTT window. As a result, the signal quality degrades below the connection threshold before the TTT expires, causing a **Radio Link Failure (RLF)**. Conversely, lowering the TTT or hysteresis parameters to trigger handovers earlier results in the **ping-pong effect**, where the UE repeatedly switches back and forth between adjacent cells, wasting network signaling capacity, increasing packet loss, and degrading Quality of Service (QoS) (as illustrated in Figure 2.1).

To resolve this trade-off, proactive prediction aims to anticipate these sudden channel drops and initiate the handover preparation phase ahead of time, using generative machine learning models to forecast RSRP trajectories before the physical channel degrades.

2.2 Review of Machine Learning Baselines

The application of Machine Learning (ML) and Deep Learning (DL) to handover management has gained significant traction. This thesis builds upon two key literature baselines.

2.2.1 Two-Stage Handover Prediction (Lima et al.)

Lima et al. [7] proposed a decoupled two-stage framework to separate trajectory forecasting from the final decision-making logic:

1. **Regression Stage:** A 4-layer LSTM network with 120 hidden units in the first layer, followed by three layers of 50 hidden units. It takes past RSRP measurements and predicts the future RSRP values over a look-ahead window. On commercial LTE drive test measurements collected in Belo Horizonte, Brazil, the LSTM regressor achieved a Mean Absolute Error (MAE) of 1.74 dB.
2. **Classification Stage:** A binary classifier (evaluating KNN, SVM, MLP, and Random Forest) takes the predicted future RSRP sequence as input and outputs a binary handover trigger decision. The Random Forest (RF) classifier achieved the highest F1-score of 97.64%, demonstrating that predicted RSRP sequences can reliably drive binary classifiers to anticipate handover triggers.

Data Imbalance Mitigation: SMOTE and Tomek Links

Because handover events (positive class) represent a very small fraction of the continuous connection time-series, the dataset is heavily skewed. To mitigate this imbalance, the authors applied a combined sampling strategy using Synthetic Minority Over-sampling Technique (SMOTE) and Tomek Links:

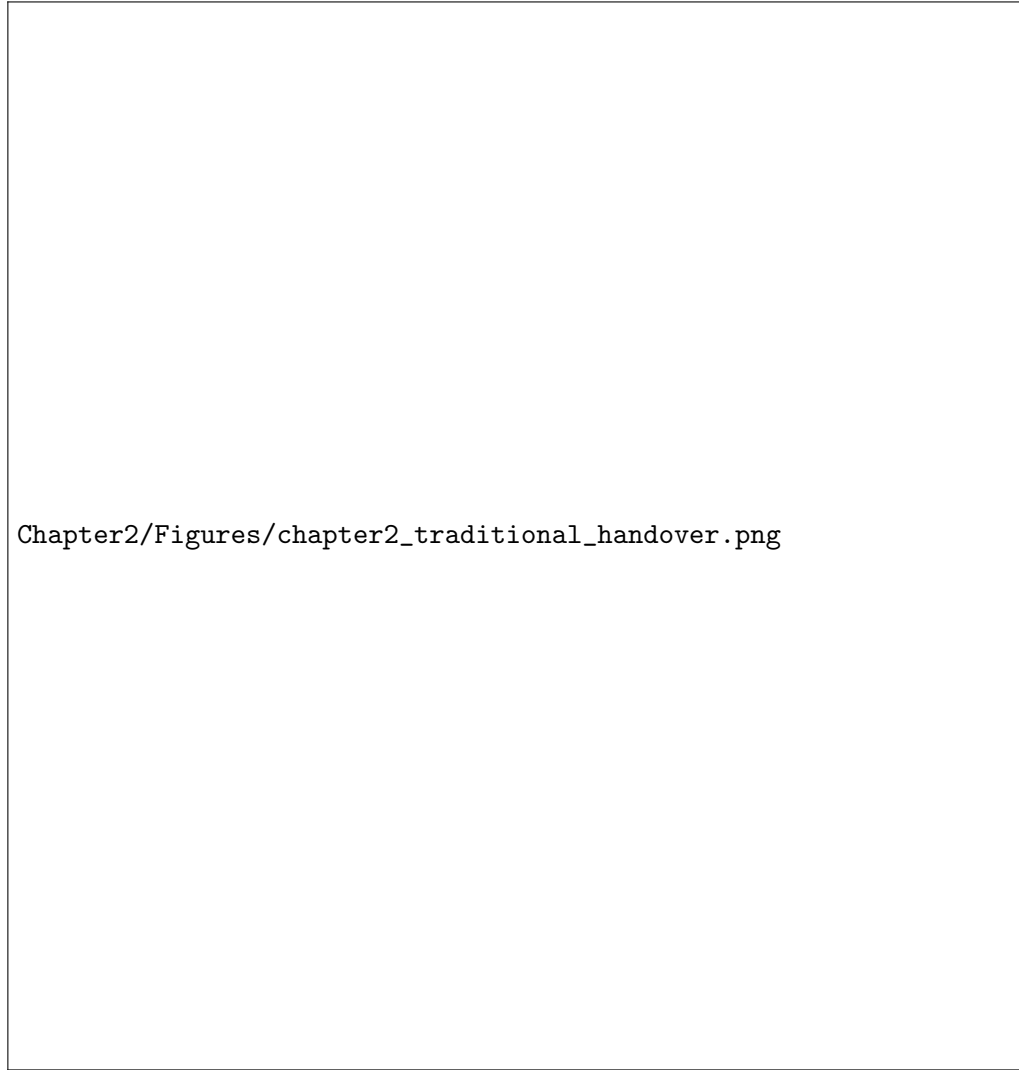


Figure 2.1: Traditional reactive handover compared with the proposed proactive handover. In (A), late handover execution due to TTT delay leads to a Radio Link Failure (RLF). In (B), the proactive model predicts the signal drop, triggering and completing the handover beforehand to maintain service continuity.

- **SMOTE:** This technique generates synthetic samples for the minority class by interpolating between existing minority examples. For each minority class sample x_i , its k -nearest neighbors within the minority class are identified. A neighbor x_{zi} is randomly selected, and a synthetic instance x_{new} is generated along the line segment joining them:

$$x_{\text{new}} = x_i + \lambda \cdot (x_{zi} - x_i) \quad (2.3)$$

where $\lambda \in [0, 1]$ is a uniform random number. This creates a denser representation of the minority class region without simply duplicating observations.

- **Tomek Links:** Once over-sampling is complete, Tomek Links are applied to remove noise and clean the decision boundaries. A Tomek Link is defined as a pair of minimally distant samples of different classes. Let x_i be a minority instance and x_j be a majority instance. The pair (x_i, x_j) forms a Tomek Link if there is no sample x_k such that:

$$d(x_i, x_k) < d(x_i, x_j) \quad \text{or} \quad d(x_j, x_k) < d(x_i, x_j) \quad (2.4)$$

where $d(\cdot, \cdot)$ denotes the Euclidean distance. Removing the majority class component of these links eliminates overlapping points near the boundary, enhancing the classifier’s performance.

While effective, this point-forecast method is deterministic. It does not provide any confidence intervals or uncertainty bounds, meaning the network must act on a single average path without knowing the risk of alternative scenarios.

2.2.2 Time-Series Benchmarking and Masking (Park et al.)

Park et al. [8] conducted a comprehensive evaluation of multiple deep learning architectures for RSRP forecasting:

- **Linear Time-Series Models (DLinear and NLinear)**: Simple yet highly effective linear architectures designed to capture temporal trends:
 - **DLinear**: This model decomposes the input sequence into a trend component X_T and a seasonal/residual component X_R using a moving average filter. It then applies separate linear layers to each component to project them to the future horizon, adding the outputs together:

$$\hat{Y} = W_T X_T + W_R X_R \quad (2.5)$$

where W_T and W_R represent the weight matrices of the linear layers.

- **NLinear**: To prevent performance degradation caused by distribution shifts (e.g., changes in mean and variance of RSRP over time), NLinear normalizes the input sequence by subtracting the last observed value x_{last} before mapping it through the linear layer, and then adds it back at the end:

$$X_{\text{norm}} = X - x_{\text{last}} \quad (2.6)$$

$$\hat{Y}_{\text{norm}} = W \cdot X_{\text{norm}} \quad (2.7)$$

$$\hat{Y} = \hat{Y}_{\text{norm}} + x_{\text{last}} \quad (2.8)$$

This simple operation ensures the model is invariant to baseline signal level changes.

- **Segment-based Recurrent Neural Networks (SegRNN)**: Rather than processing the time-series step-by-step (which suffers from vanishing gradients over long horizons), SegRNN segmentizes the input sequence into non-overlapping blocks of length L . An RNN is run over these segment embeddings, reducing the recurrent path length by a factor of L and significantly accelerating execution speed while preserving temporal dependencies.
- **Transformer Models (Informer and Autoformer)**: These models replace standard attention with sparse attention mechanisms (ProbSparse attention in Informer and auto-correlation in Autoformer) to reduce the sequence length computational complexity from $O(L^2)$ to $O(L \log L)$.

The evaluation demonstrated that linear models (DLinear/NLinear) and SegRNN models achieved competitive Top-1 Margin Accuracy, proving that modern point-forecasting architectures can capture temporal trends. However, like the two-stage framework proposed by Lima et al. [7], all of these architectures remain deterministic.

2.3 Generative Denoising Diffusion Theory

To overcome the limitations of deterministic point forecasting, this thesis utilizes generative diffusion models. Diffusion models are a class of probabilistic generative models that learn a target distribution by reversing a gradual noise-injection process.

2.3.1 Denoising Diffusion Probabilistic Models (DDPM)

A standard Denoising Diffusion Probabilistic Model (DDPM) [12, 13] consists of two main phases: a forward diffusion process that systematically destroys data structure, and a reverse denoising process that learns to reconstruct the data.

Forward Diffusion Process

The forward process is a Markov chain that gradually adds Gaussian noise to the clean data $x_0 \sim q(x)$ over T steps according to a variance schedule $\beta_1, \beta_2, \dots, \beta_T$:

$$q(x_1, \dots, x_T | x_0) = \prod_{t=1}^T q(x_t | x_{t-1}) \quad (2.9)$$

Here, $q(x_1, \dots, x_T | x_0)$ represents the joint probability distribution of the forward diffusion trajectory $x_1, \dots, x_T$ conditioned on the original clean signal x_0 . The variable x_0 is the initial clean sample from the data distribution (representing the true future RSRP trajectory), x_t represents the noisy intermediate state of the data at diffusion step t , T is the total number of diffusion steps (diffusion time horizon), and $\prod_{t=1}^T$ denotes the product of transition probabilities across all steps.

Each individual transition is parameterised as:

$$q(x_t | x_{t-1}) = \mathcal{N}\left(x_t; \sqrt{1 - \beta_t} x_{t-1}, \beta_t \mathbf{I}\right) \quad (2.10)$$

where $\mathcal{N}(\cdot; \mu, \Sigma)$ represents the multivariate Gaussian (normal) distribution with mean vector μ and covariance matrix Σ . The parameter $\beta_t \in (0, 1)$ is the variance scheduler value at step t , which controls the amount of noise added at each step, and $\mathbf{I}$ is the identity matrix representing isotropic noise covariance.

Using the helper variables $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$, we can bypass the recursive steps and express the distribution of x_t at any arbitrary step t directly in closed form conditioned on the original sample x_0 :

$$q(x_t | x_0) = \mathcal{N}\left(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) \mathbf{I}\right) \quad (2.11)$$

Here, $\alpha_t = 1 - \beta_t$ is the fraction of the previous signal preserved at step t , and $\bar{\alpha}_t$ is the cumulative signal preservation coefficient up to step t .

Thus, a noisy sample x_t can be expressed as:

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon \quad \text{where } \epsilon \sim \mathcal{N}(0, \mathbf{I}) \quad (2.12)$$

where ϵ represents a standard Gaussian noise vector sampled from $\mathcal{N}(0, \mathbf{I})$. As t approaches T , the variance parameter $\bar{\alpha}_t$ goes to 0, and the data distribution is completely destroyed, leaving x_T as standard isotropic Gaussian noise.

Reverse Denoising Process

The reverse process is defined as a joint distribution $p_\theta(x_0, \dots, x_T)$ that starts from a standard Gaussian prior $p(x_T) = \mathcal{N}(x_T; 0, \mathbf{I})$ and runs a parameterized Markov chain in reverse:

$$p_\theta(x_{0:T}) = p(x_T) \prod_{t=1}^T p_\theta(x_{t-1} | x_t) \quad (2.13)$$

where $p_\theta(x_{0:T})$ is the joint probability of the reverse denoising trajectory from step T down to the clean signal x_0 , and $p(x_T)$ is the prior isotropic Gaussian noise distribution.

The individual transitions are parameterised as Gaussian distributions:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \quad (2.14)$$

where $\mu_\theta(x_t, t)$ represents the predicted mean vector of the transition and $\Sigma_\theta(x_t, t)$ represents its covariance matrix.

The neural network is trained to predict the parameters of these transitions. In the formulation by standard DDPM papers [12, 13], the covariance is fixed to a time-dependent schedule $\Sigma_\theta(x_t, t) = \sigma_t^2 \mathbf{I}$, where the scalar variance parameter σ_t^2 is set to either β_t or the posterior variance $\tilde{\beta}_t$:

$$\tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t \quad (2.15)$$

where $\tilde{\beta}_t$ represents the mathematically derived analytical posterior variance of the reverse process when conditioned on the clean sample x_0 .

The mean vector $\mu_\theta(x_t, t)$ is parameterized to predict the noise ϵ added during the forward process:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right) \quad (2.16)$$

where $\epsilon_\theta(x_t, t)$ represents the output of the neural network (specifically, our conditional U-Net structure) that predicts the noise component at step t .

This parameterization allows the model to be trained using a simplified Mean Squared Error (MSE) objective:

$$L_{\text{simple}}(\theta) = \mathbb{E}_{t, x_0, \epsilon} [\|\epsilon - \epsilon_\theta(x_t, t)\|^2] \quad (2.17)$$

where θ represents the learnable parameters (weights and biases) of the neural network, $L_{\text{simple}}(\theta)$ is the simplified training loss, t is sampled uniformly from $[1, T]$, x_0 is sampled from the training dataset, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ is the true injected noise, and $\mathbb{E}_{t, x_0, \epsilon}$ denotes the mathematical expectation operator over these variables.

Score Matching Perspective

An alternative framework for understanding diffusion models is score matching. The score function of a probability density $p(x)$ is defined as the gradient of its log-density with respect to the data: $\nabla_x \log p(x)$. In denoising diffusion models, the noise predictor network $\epsilon_\theta(x_t, t)$ is mathematically related to the score function of the noise-perturbed data distribution $q(x_t)$ at scale t :

$$\nabla_{x_t} \log q(x_t) \approx -\frac{\epsilon_\theta(x_t, t)}{\sqrt{1 - \bar{\alpha}_t}} \quad (2.18)$$

Here, $\nabla_{x_t} \log q(x_t)$ represents the score function (the gradient of the log-probability density of the noisy data x_t with respect to the noisy data), $\epsilon_\theta(x_t, t)$ represents the neural network noise predictor parameterized by θ , and $1 - \bar{\alpha}_t$ is the noise scaling coefficient. By training the network to predict the injected noise, the model implicitly learns the gradient vectors that point toward regions of higher probability density, allowing it to denoise arbitrary starting states by following the score field.

2.3.2 Denoising Diffusion Implicit Models (DDIM)

Standard DDPM sampling is stochastic because it injects random Gaussian noise at each reverse step t . Furthermore, the Markovian assumption forces the sampler to trace all T training steps, which is computationally expensive.

Denoising Diffusion Implicit Models (DDIM) [14] generalise this by proposing a non-Markovian family of forward processes. For a given noise schedule, this formulation shares the same

marginal distributions $q(x_t|x_0)$ as DDPM, allowing us to reuse the same trained network weights θ . The reverse process transition is parameterized as:

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\hat{x}_0(x_t, \epsilon_\theta) + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}\epsilon_\theta(x_t, t) + \sigma_t\epsilon_t \quad (2.19)$$

where x_{t-1} is the reconstructed intermediate sample at the previous step $t - 1$ in the sub-sequence, $\bar{\alpha}_{t-1}$ is the cumulative signal preservation coefficient at step $t - 1$, $\hat{x}_0(x_t, \epsilon_\theta)$ is the predicted clean sample at step 0 reconstructed from the current noisy state x_t , the scaling parameter σ_t controls the stochasticity of the reverse step, $\epsilon_\theta(x_t, t)$ is the noise predicted by the neural network, and $\epsilon_t \sim \mathcal{N}(0, \mathbf{I})$ is standard Gaussian noise vector added at step t .

The predicted clean sample $\hat{x}_0$ is defined as:

$$\hat{x}_0(x_t, \epsilon_\theta) = \frac{x_t - \sqrt{1 - \bar{\alpha}_t}\epsilon_\theta(x_t, t)}{\sqrt{\bar{\alpha}_t}} \quad (2.20)$$

where $\bar{\alpha}_t$ is the cumulative signal preservation coefficient at step t , and x_t is the noisy sample at step t .

The parameter σ_t controls the stochasticity of the reverse process. By setting $\sigma_t = 0$ for all t , the process becomes entirely deterministic, meaning x_{t-1} is computed algebraically from x_t and the predicted noise. This allows for sub-sequence sampling: we can define a subset of steps $S < T$ (e.g., $S = 50$) and skip intermediate steps, reducing inference time significantly while maintaining sample diversity through the initial noise vector x_T .

Methodology

Preface

In this chapter, we detail our proposed proactive handover prediction methodology. We formalize RSRP forecasting as a conditional probabilistic time-series generation problem. We then present the network architecture of our RSRP-Diffusion model and explain the deterministic DDIM sampler speedup.

3.1 Problem Formulation and Decision Logic

Let the historical window of RSRP measurements collected from a serving cell be denoted as:

$$X_{\text{hist}} = [x_{t-W_{\text{in}}+1}, \dots, x_t] \in \mathbb{R}^{W_{\text{in}} \times 1} \quad (3.1)$$

where X_{hist} is the history matrix of observed RSRP values, and W_{in} is the history sequence length (configured to 50 ms, representing a sequence of 50 samples at a 1 ms sampling interval). Our objective is to predict the future RSRP trajectory:

$$Y_{\text{future}} = [y_{t+1}, \dots, y_{t+W_{\text{out}}}] \in \mathbb{R}^{W_{\text{out}} \times 1} \quad (3.2)$$

where Y_{future} is the true future RSRP trajectory vector, and W_{out} is the look-ahead forecasting horizon. For this short-horizon configuration, W_{out} is set to 10 ms (representing 10 future steps). This specific horizon is selected to align with the physical 5G radio frame duration (10 ms), acting as the minimum boundary for high-frequency channel feedback. Over this short window, the signal behaves linearly, allowing parallel trajectory generation to perform accurately without decay, while keeping execution time within the strict sub-200 ms SDN control loop budget.

Rather than outputting a single point estimate (like deterministic LSTMs), our generative models produce an ensemble of N probabilistic future trajectories:

$$\{Y^{(s)}\}_{s=1}^N \quad \text{where} \quad Y^{(s)} = [y_{t+1}^{(s)}, \dots, y_{t+W_{\text{out}}}^{(s)}] \in \mathbb{R}^{W_{\text{out}} \times 1} \quad (3.3)$$

where $\{Y^{(s)}\}_{s=1}^N$ represents the set of generated future trajectories, $Y^{(s)}$ is the s -th simulated trajectory vector, and N represents the ensemble sample size.

Here, the sample size is configured to $N = 50$. The choice of exactly 50 samples is a critical trade-off between statistical accuracy and computational complexity. A smaller ensemble (e.g., $N = 10$ or 20) introduces high variance and fails to resolve the tail risk of signal collapse accurately. A larger ensemble (e.g., $N = 100$) provides only marginal improvements in failure probability estimation but increases the GPU/CPU inference latency linearly, violating real-time execution constraints of Near-RT RIC systems. Generating $N = 50$ trajectories provides a robust probability resolution of 2% while remaining within the tight millisecond-level execution budget. This ensemble represents the conditional probability density function $P(Y_{\text{future}}|X_{\text{hist}})$.

3.1.1 Physical Context of RSRP Thresholds

In commercial mobile networks, the Reference Signal Received Power (RSRP) measures the average power of the resource elements that carry cell-specific reference signals. The receiver sensitivity of typical user equipment is bounded. A signal level of -85 dBm represents a critical boundary in high-frequency (mmWave/THz) bands:

- **Normal Coverage** (> -80 dBm): High signal quality, low block error rates (BLER), and support for high-order modulation schemes (e.g., 64-QAM or 256-QAM).
- **Edge Coverage** (between -80 and -90 dBm): Degraded signal quality. The system falls back to lower modulation schemes (e.g., QPSK) and incurs higher retransmission rates.
- **Outage Coverage** (< -95 dBm): Highly vulnerable to connection dropouts. At these levels, the probability of decoding control messages drops significantly, leading to Radio Link Failure (RLF).

We set the failure threshold $\gamma = -85$ dBm as a proactive trigger level, which is high enough to allow signaling time before the physical channel completely collapses to -95 dBm.

3.1.2 Risk Assessment and Handover Trigger Logic

For each simulated future trajectory sample $Y^{(s)}$, we evaluate whether the signal power is predicted to cross the critical failure threshold γ :

$$\mathbb{I}^{(s)} = \mathbb{I} \left(\min_{k=1 \dots W_{\text{out}}} y_{t+k}^{(s)} < \gamma \right) \quad (3.4)$$

where $\mathbb{I}$ is the binary indicator function. The failure probability (or risk factor) P_{failure} is then computed as the proportion of simulated futures that cross this threshold:

$$P_{\text{failure}} = \frac{1}{N} \sum_{s=1}^N \mathbb{I}^{(s)} \quad (3.5)$$

Based on this computed risk P_{failure} , the network controller triggers one of three control actions:

1. **STAY CONNECTED** ($P_{\text{failure}} \leq 0.4$): The serving signal is predicted to remain stable with high confidence. The connection is maintained without any signaling overhead.
2. **PREPARE HANDOVER** ($0.4 < P_{\text{failure}} \leq 0.8$): The serving signal shows a moderate risk of failure. The controller proactively coordinates with candidate neighbor cells to prepare radio resources and configure measurement gaps.
3. **TRIGGER HANDOVER IMMEDIATELY** ($P_{\text{failure}} > 0.8$): The serving signal is highly likely to fail. The controller immediately issues a Handover Command to redirect the user's connection to the best candidate cell, bypassing the standard Time-To-Trigger delay to avoid a Radio Link Failure.

3.1.3 Advantages of Probabilistic Decision Making

To highlight the advantages of this probabilistic approach, consider a scenario where the true RSRP behaves randomly due to shadowing. If a deterministic model predicts a single future RSRP of -83 dBm (which is above the failure threshold $\gamma = -85$ dBm), the network controller would take no action. However, if the actual distribution is wide, there could be a 40% probability that the signal falls to -88 dBm.

Under our probabilistic framework, this scenario yields $P_{\text{failure}} = 0.40$, triggering the ****PREPARE HANDOVER**** phase. If the signal indeed drops, the target cell is already prepared, and the handover executes immediately, preventing an RLF. If the signal recovers, the preparation is cancelled, incurring only a minor signaling overhead. This demonstrates how probabilistic forecasts allow the network to manage risk dynamically.

3.2 RSRPDiffusion (DDPM) Architecture

The proposed primary architecture, **RSRPDiffusion**, is a conditional Denoising Diffusion Probabilistic Model (DDPM) designed to predict future RSRP trajectories. The network takes three inputs: the noisy future RSRP trajectory $x_{\text{future}}^{\text{noisy}} \in \mathbb{R}^{W_{\text{out}} \times 1}$, the current diffusion timestep $t \in \mathbb{R}$, and the historical context $x_{\text{history}} \in \mathbb{R}^{W_{\text{in}} \times 1}$. The target of the model is to predict the noise injected into the future trajectory, which is subsequently subtracted during the reverse sampling steps.

The internal structure consists of the following components:

1. **Context RNN (History Encoder)**: A single-layer Gated Recurrent Unit (GRU) with a hidden dimension of 64 is utilized to model the sequential dependencies in the historical

RSRP observations. It processes the 50 ms history sequence step-by-step and extracts the final hidden state vector $h_d[-1] \in \mathbb{R}^{64}$ as a compact representation of recent mobility trends:

$$\text{context} = \text{GRU}(x_{\text{history}}) \quad (3.6)$$

The GRU cell state is updated recursively at each history step j using the standard gating equations:

$$z_j = \sigma(W_z x_j + U_z h_{j-1} + b_z) \quad (3.7)$$

$$r_j = \sigma(W_r x_j + U_r h_{j-1} + b_r) \quad (3.8)$$

$$\tilde{h}_j = \tanh(W_h x_j + U_h (r_j \odot h_{j-1}) + b_h) \quad (3.9)$$

$$h_j = (1 - z_j) \odot h_{j-1} + z_j \odot \tilde{h}_j \quad (3.10)$$

where z_j is the update gate, r_j is the reset gate, and h_j is the hidden state. This recurrent filtering extracts macro-mobility trends while smoothing out micro-fading channel fluctuations.

2. **Sinusoidal Time Embedding:** The diffusion timestep $t \in [1, T]$ (where $T = 100$) is embedded into a dense vector space using Sinusoidal Position Embeddings. This maps the scalar integer t to a coordinate representation of dimension $D = 128$, allowing the network to distinguish the noise level at different diffusion steps. For each index $i \in [0, D/2 - 1]$, the embedding components are defined as:

$$\text{Embed}(t)_{2i} = \sin\left(t \cdot 10000^{-\frac{2i}{D}}\right) \quad (3.11)$$

$$\text{Embed}(t)_{2i+1} = \cos\left(t \cdot 10000^{-\frac{2i}{D}}\right) \quad (3.12)$$

This time embedding is then projected through a Time MLP to align it with the network state:

$$t_{\text{emb}} = \text{Linear}(\text{SiLU}(\text{Linear}(\text{Embed}(t)))) \in \mathbb{R}^{128} \quad (3.13)$$

where SiLU is the Sigmoid Linear Unit activation function, defined mathematically as:

$$\text{SiLU}(x) = x \cdot \sigma(x) = \frac{x}{1 + e^{-x}} \quad (3.14)$$

3. **Input and Context Projections:** The noisy future sequence $x_{\text{future}}^{\text{noisy}}$ is projected to the hidden dimension $D = 128$ via a linear layer. Similarly, the GRU context vector is projected to the same dimension:

$$x_{\text{proj}} = \text{Linear}(x_{\text{future}}^{\text{noisy}}) \quad (3.15)$$

$$c_{\text{proj}} = \text{Linear}(\text{context}) \quad (3.16)$$

These projected representations are combined element-wise along with the time embedding:

$$h_0 = x_{\text{proj}} + t_{\text{emb}} + c_{\text{proj}} \quad (3.17)$$

where the time embedding and context projections are unsqueezed to match the sequence dimensions of x_{proj} . Element-wise addition is selected instead of concatenation to keep the internal tensor dimension constant at $D = 128$, which reduces the weight parameter count of subsequent layers and speeds up model inference.

4. **Residual Denoising Blocks:** The combined state h_0 passes through a series of 4 Residual Blocks. Each block contains two linear layers with SiLU activations and a dropout rate of 0.1:

$$h_{l+1} = h_l + \text{Dropout}(\text{Linear}(\text{SiLU}(\text{Linear}(h_l)))) \quad (3.18)$$

The skip connection ensures stable gradient flow during backpropagation and prevents vanishing gradients across the deep linear layers.

5. **Output Projection:** Finally, a linear layer projects the state back to the original input dimension (1) to output the predicted noise:

$$\hat{\epsilon} = \text{Linear}(h_L) \quad (3.19)$$

The model is optimized using Mean Squared Error (MSE) between the actual noise ϵ and the predicted noise $\hat{\epsilon}$:

$$L_{\text{MSE}} = \frac{1}{W_{\text{out}}} \sum_{k=1}^{W_{\text{out}}} \|\epsilon_k - \hat{\epsilon}_k\|^2 \quad (3.20)$$

This parallel formulation allows the model to predict the noise for all W_{out} future steps concurrently, ensuring fast parallel generation during inference. The forward and reverse diffusion processes for trajectory generation are illustrated in Figure 3.2. The internal neural network layers and detailed tensor transformations of the proposed **RSRPDiffusion** model architecture are shown in Figure 3.1.

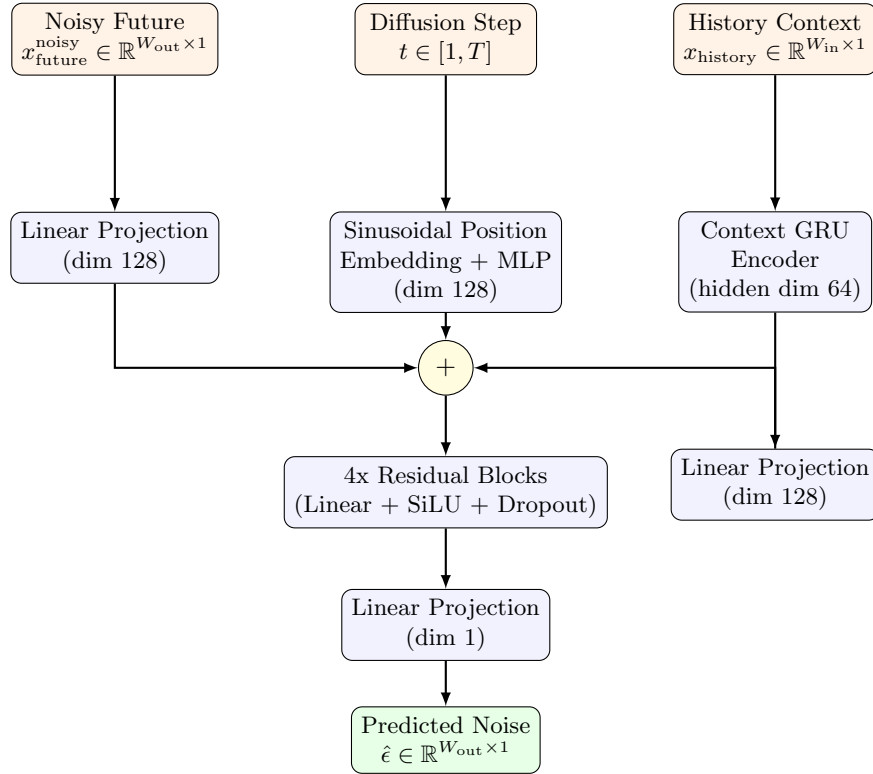


Figure 3.1: Detailed block diagram of the proposed **RSRPDiffusion** neural network architecture, showing input processing, context recurrent encoding, temporal embeddings, element-wise addition, residual processing backbone, and noise output projection.

3.3 DDIM Inference Optimization

Although the conditional DDPM model provides high-quality probabilistic forecasts, running $T = 100$ denoising steps in parallel for each sample introduces significant computational latency, which can violate the real-time requirements of 5G network control loops. To address this, we implement Denoising Diffusion Implicit Models (DDIM) sampling.

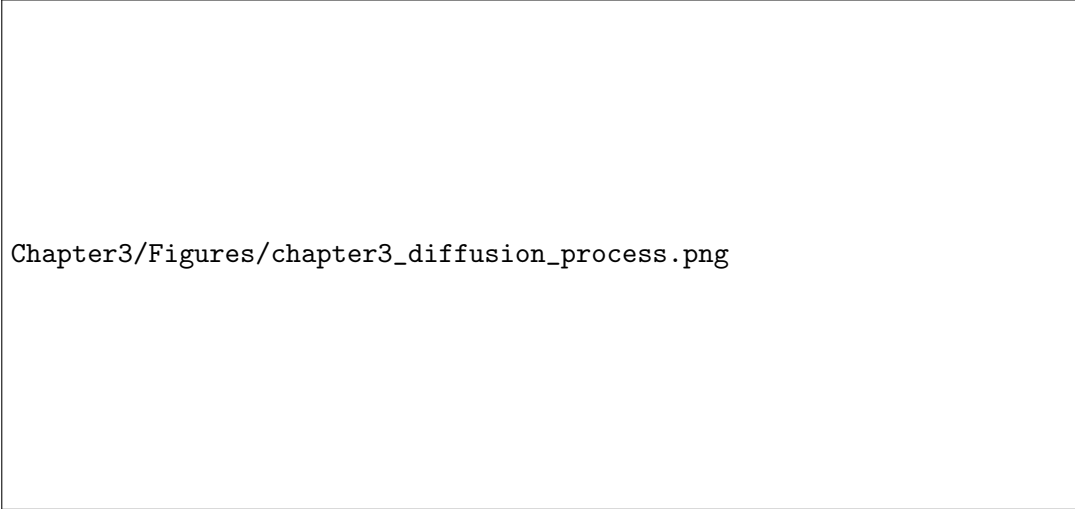


Figure 3.2: Forward diffusion process (noise addition) and reverse conditional denoising process (trajectory generation) for RSRP forecasting.

3.3.1 Deterministic Denoising Formulation

DDIM [14] generalizes the generative reverse process to a class of non-Markovian decision paths. By setting the stochasticity parameter $\eta = 0$, the reverse process becomes entirely deterministic. Given a noisy future trajectory x_t at step t , the predicted clean trajectory x_0 is first reconstructed and clamped for numerical stability:

$$\hat{x}_0 = \text{clamp} \left(\frac{x_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(x_t, t, h)}{\sqrt{\bar{\alpha}_t}}, -1.0, 1.0 \right) \quad (3.21)$$

The representation at the previous step x_{prev} is then computed as:

$$x_{\text{prev}} = \sqrt{\bar{\alpha}_{\text{prev}}} \hat{x}_0 + \sqrt{1 - \bar{\alpha}_{\text{prev}}} \epsilon_\theta(x_t, t, h) \quad (3.22)$$

where prev denotes the preceding timestep in the sub-sequence. Because this process is deterministic, the variation in generated scenarios is driven entirely by the initial Gaussian noise sample $x_T \sim \mathcal{N}(0, \mathbf{I})$ rather than stochastic denoising steps. This deterministic mapping allows us to sample the space of future RSRP trajectories with high consistency.

Role of Physical Clamping in RSRP Forecasting

The inclusion of the `clamp` operation in the reconstruction of $\hat{x}_0$ is highly critical for RSRP time-series forecasting. RSRP values in our dataset are normalized to the range $[-1.0, 1.0]$. During the initial stages of the denoising process (where t is large and noise dominates), the model’s prediction of $\hat{x}_0$ can experience extreme values (e.g., < -2.5 or > 2.5) due to noise amplification.

Without clamping, these out-of-bound values propagate recursively through the reverse process, leading to gradient explosion, numerical overflow, and divergence of the predicted RSRP. By clamping $\hat{x}_0$ to $[-1.0, 1.0]$ at each step, we enforce a physical boundary constraint corresponding to the RSRP bounds (-101.0 dBm to -45.5 dBm), keeping the trajectories bounded and physically realistic.

3.3.2 Sub-sequence Step Scheduling

DDIM allows us to define a subset of timesteps $S < T$ for the reverse path. We define a sub-sequence of length $S = 50$ timesteps by evenly sampling the original 100 training timesteps

in reverse order:

$$\tau = [\tau_1, \tau_2, \dots, \tau_S] \subset [0, 99] \quad (3.23)$$

where $\tau_i = 99 - (i - 1) \times 2$ (representing the sequence $[99, 97, 95, \dots, 1]$).

By running the reverse process only on this sub-sequence, the number of neural network evaluations per window is halved from 100 to 50, achieving a $2\times$ inference speedup. This optimization requires no retraining and uses the same checkpoint weights as the trained DDPM model. While more complex schedules (such as quadratic spacing, which places more steps near $t = 0$) are common in image generation, we find that uniform linear spacing provides optimal and stable results for 1D RSRP time-series forecasting.

Experimental Setup

Preface

In this chapter, we outline our experimental setup and evaluation methodology. We describe the commercial LTE drive-test dataset, our preprocessing and normalization protocols, and present the mathematical definitions of the evaluation metrics used to assess our model’s performance: Mean Absolute Error (MAE), Continuous Ranked Probability Score (CRPS), False Negative Rate (FNR), and inference latency.

4.1 Dataset Description and Preprocessing

To validate our proposed proactive handover prediction methodology, we utilize real-world drive test measurements collected from a commercial LTE network in Belo Horizonte, Brazil. This is the exact dataset used in the baseline work by Lima et al. [7], ensuring a fair and consistent comparison.

4.1.1 Data Collection Campaign and Urban Environment

The measurements were collected using a Rohde & Schwarz TSMW radio network analyzer mounted on a vehicle traveling at an average speed of approximately 50 km/h in an urban environment. RSRP signals were sampled at high-frequency intervals of 1 ms. The data collection campaign was conducted in downtown Belo Horizonte, characterized by high-rise building canyons, dense pedestrian traffic, and varied street layouts. These topographical features create significant multipath fading, diffraction, and sudden signal blockages, making this dataset highly representative of the challenging channel propagation environments typical of high-frequency mmWave deployments.

The dataset is divided into three continuous segments, saved as CSV files:

- `drive_test_measurements01.csv`: 120,000 samples (training set)
- `drive_test_measurements02.csv`: 120,000 samples (training set)
- `drive_test_measurements03.csv`: 60,000 samples (held-out test set, never seen during model training or hyperparameter tuning)

Additionally, a 3GPP-compliant system-level simulator and an ns-3 network simulator are utilized to model the multi-cell network layout and generate trajectory pathways for UEs moving at 50–60 km/h, as shown in Figure 4.1.

4.1.2 Preprocessing and Normalization

The raw training RSRP measurements range from a minimum of $x_{\min} = -101.0$ dBm to a maximum of $x_{\max} = -45.5$ dBm. In radio communication engineering:

- The lower bound of -101.0 dBm corresponds to the receiver sensitivity noise floor of the radio network analyzer for the configured channel bandwidth. Signals below this level are completely lost in thermal noise.
- The upper bound of -45.5 dBm represents close physical proximity (typically within 50–100 meters) to the base station sector antenna under Line-of-Sight conditions.

For the diffusion process to converge stably, RSRP values are normalized to the range $[-1, 1]$ using min-max scaling:

$$x_{\text{normalized}} = \frac{x_{\text{dBm}} - x_{\min}}{x_{\max} - x_{\min}} \times 2 - 1 \quad (4.1)$$

The normalization statistics ($x_{\min}$ and $x_{\max}$) are computed exclusively from the training dataset and subsequently applied to the held-out test dataset to prevent data leakage.

4.1.3 Windowing Protocol

The continuous time-series data is segmented using a sliding window protocol with a step size of 1 ms:

- **History Window (W_{in})**: The model takes the past 50 steps (50 ms of history) to establish the current mobility context.

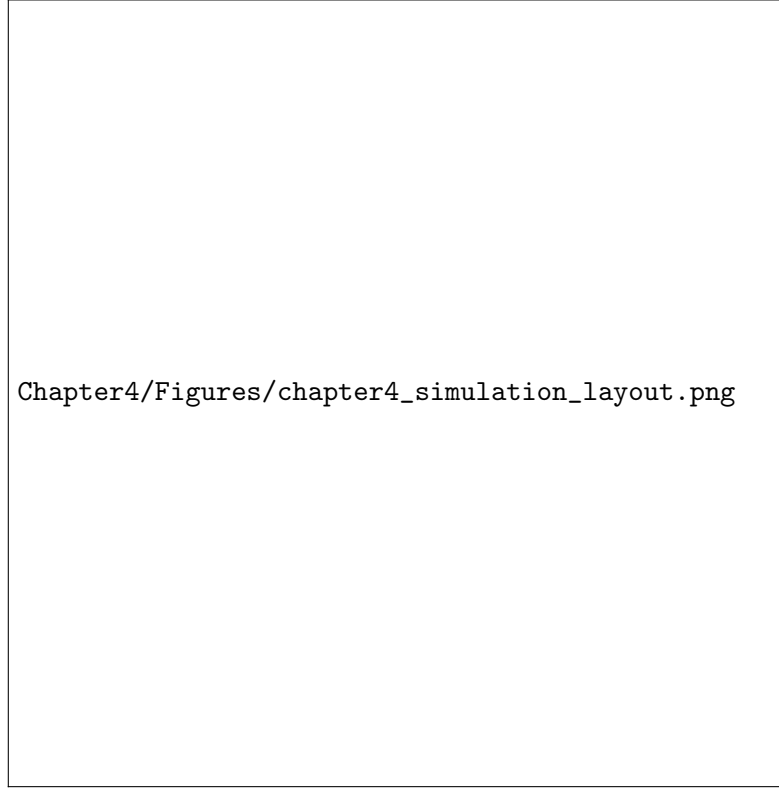


Figure 4.1: 3GPP-compliant multi-cell hexagonal network layout with cell sectors, base stations (gNBs), and the simulated User Equipment (UE) trajectory path.

- **Prediction Window (W_{out}):** The model predicts future trajectories over a short horizon of 10 steps (10 ms) or a long horizon of 50 steps (50 ms).

This short look-ahead window is aligned with the real-time constraints of SDN control loops, which operate within sub-second time budgets, enabling proactive routing table reconfigurations.

4.2 Evaluation Metrics

To evaluate the predictive accuracy, probabilistic calibration, signaling safety, and computational speed of the models, we define four core metrics.

4.2.1 Mean Absolute Error (MAE)

The accuracy of the mean RSRP prediction is measured in decibels (dB) via the Mean Absolute Error:

$$\text{MAE} = \frac{1}{W_{\text{out}}} \sum_{k=1}^{W_{\text{out}}} |y_{t+k} - \hat{y}_{t+k}| \quad (4.2)$$

where y_{t+k} is the ground truth RSRP and $\hat{y}_{t+k} = \frac{1}{N} \sum_{s=1}^N \hat{y}_{t+k}^{(s)}$ is the mean predicted RSRP across the N generated future trajectories.

4.2.2 Continuous Ranked Probability Score (CRPS)

To evaluate the quality of the predicted probability distribution (evaluating both its accuracy and calibration), we use the Continuous Ranked Probability Score (CRPS). Standard metrics

like MSE or MAE only evaluate point predictions, ignoring the model's confidence or variance. CRPS resolves this by measuring the distance between the predicted cumulative distribution function (CDF) F and the empirical CDF of the observation y :

$$\text{CRPS}(F, y) = \int_{-\infty}^{\infty} (F(z) - \mathbb{I}(z \geq y))^2 dz \quad (4.3)$$

where $\mathbb{I}$ is the indicator function, which equals 1 if $z \geq y$ and 0 otherwise.

For an ensemble of generated predictions $\{y^{(s)}\}_{s=1}^N$ (in our case, $N = 50$), the CRPS can be decomposed into an accuracy term and a diversity (spread) term:

$$\text{CRPS} = \frac{1}{N} \sum_{s=1}^N |y^{(s)} - y| - \frac{1}{2N^2} \sum_{s=1}^N \sum_{s'=1}^N |y^{(s)} - y^{(s')}| \quad (4.4)$$

The first term measures the average absolute error of the forecasts, while the second term rewards the diversity (spread) of the trajectories, penalizing overconfident (under-dispersed) or underconfident (over-dispersed) distributions. A lower CRPS value indicates a better calibrated distribution (with $\text{CRPS} = 0$ being perfect). This metric ensures that the generated distribution reflects the true physical uncertainty of the channel.

4.2.3 False Negative Rate (FNR)

For network safety, a missed failure is critical because it means the serving signal degraded below the connection threshold without triggering handover preparation, likely resulting in a connection drop. The False Negative Rate (FNR) is defined as:

$$\text{FNR} = \frac{\text{Missed Failures}}{\text{Total Actual Failures}} \times 100\% \quad (4.5)$$

where:

- An actual failure is recorded if the real RSRP drops below $\gamma = -85$ dBm at any point in the prediction window W_{out} .
- A missed failure occurs if the actual signal failed but the model predicted a failure probability $P_{\text{failure}} \leq 0.4$ (meaning the controller took the action and failed to trigger any handover preparation).

A low FNR is highly critical: while a false positive (predicting a failure that does not happen) only incurs a temporary resource allocation overhead on target gNBs, a false negative leads to service disruption and dropped calls.

4.2.4 Inference Latency and Speedup

Inference latency measures the average wall-clock time (in seconds) required to generate $N = 50$ future trajectories for a single 50 ms history window. The speedup factor relative to the baseline DDPM is defined as:

$$\text{Speedup} = \frac{\text{Inference Time}_{\text{DDPM}}}{\text{Inference Time}_{\text{Model}}} \quad (4.6)$$

To understand the physical significance of latency, consider a vehicle traveling at 50 km/h (≈ 13.9 m/s or 1.4 cm per millisecond). During a 10 ms forecast window, the vehicle travels 14 cm. In a dense mmWave urban environment, a signal blockage can occur within a few centimeters of movement. Thus, the combined inference and decision execution latency must remain well below the physical channel degradation window to allow the SDN controller to push flow rules before the connection drops.

Results & Analysis

Preface

In this chapter, we present the quantitative evaluation of the models. We compare parallel DDPM and fast DDIM forecasting across a 10 ms look-ahead window. We analyze RSRP prediction errors, continuous ranked probability scores, false negative rates, and computational latencies. We also examine the calibration curves and rolling stability of the generated distributions.

5.1 Experimental Results: Short Prediction Horizon (10 ms)

We evaluate the performance of the parallel diffusion model configurations over a short prediction horizon ($W_{\text{out}} = 10$ steps, corresponding to a 10 ms look-ahead window):

1. **DDPM-10**: Standard conditional parallel diffusion (100 denoising steps).
2. **DDIM-10**: Deterministic parallel diffusion (50 denoising steps, utilizing the same trained DDPM weights).

Table 5.1 reports the quantitative comparison of these configurations evaluated on 100 sliding windows from the held-out test dataset.

Table 5.1: Model Performance Comparison for Group 1 (10 ms Horizon)

Model	MAE (dB)	CRPS	FNR (%)	Time/Window (s)	Speedup
DDPM-10 (Baseline)	3.9214	2.8218	21.43%	0.3712	1.00×
DDIM-10 (Fast)	4.0362	2.8626	14.29%	0.1887	1.97×

5.1.1 Analysis of Performance and Latency Trade-offs

As reported in Table 5.1, **DDIM-10** (running only 50 denoising steps) achieves the fastest execution time of 0.1887 seconds per window, representing a **1.97×** **speedup** over the baseline DDPM-10. This speedup is achieved with almost no loss in forecasting quality: the MAE increases by only 0.11 dB (from 3.9214 dB to 4.0362 dB), and the CRPS increases by only 0.04. This demonstrates that the deterministic reverse path of DDIM allows us to halve the number of denoising steps without sacrificing the statistical quality of the predictions.

5.1.2 Statistical Evaluation of False Negative Rates

Crucially, DDIM-10 achieves the lowest False Negative Rate of **14.29%**, compared to 21.43% for DDPM-10. This result is highly important for network deployment safety. In standard DDPM, the reverse denoising path is stochastic, adding random Gaussian noise $\sigma_t z_t$ at each step. This stochasticity increases the variance of the generated trajectories, occasionally generating optimistic paths that do not cross the failure threshold $\gamma = -85$ dBm during a blockage event, causing a missed trigger.

By setting the stochasticity parameter $\eta = 0$, DDIM-10 removes this random noise injection during sampling. The variation in the generated trajectories is driven entirely by the initial noise vectors $x_T \sim \mathcal{N}(0, \mathbf{I})$ mapped through a deterministic trajectory. This deterministic mapping produces a more concentrated and stable distribution that covers the tail risk of signal drops more consistently, reducing the rate of missed triggers.

The physical threshold of $\gamma = -85$ dBm represents the boundary between edge coverage and complete connection outage. In standard commercial user equipment, below -85 dBm, the receiver’s Block Error Rate (BLER) starts to rise exponentially. If the signal continues to drop and falls below -95 dBm, the physical link completely collapses, resulting in a Radio Link Failure (RLF). By setting the proactive decision threshold at $\gamma = -85$ dBm, the model provides a 10 dB safety buffer zone. This buffer allows the SDN controller sufficient time to complete signaling and resource reservation at the target base station before the terminal experiences a catastrophic connection loss. This makes DDIM-10 not only the fastest model but also the safest for critical mobility management.

5.2 Analysis of Visualizations and Calibration

We analyze the thesis-quality figures generated during our experimental evaluation, which are stored in the `Chapter5/Figures/` directory. These visualizations illustrate the performance, error distribution, probability calibration, and temporal stability of the proposed diffusion models.

5.2.1 Metrics and Error Distributions

Figure 5.1 summarizes the key evaluation metrics (MAE, CRPS, FNR, and Inference Time) side-by-side for DDPM and DDIM. It visually highlights the substantial speedup of DDIM compared to the baseline DDPM.

Chapter5/Figures/fig1_metrics_bar.png

Figure 5.1: Comparison of key evaluation metrics (MAE, CRPS, FNR, and Inference Time) for DDPM and DDIM-10. DDIM-50 reduces inference time by half compared to DDPM-10 with minimal accuracy loss.

To understand the dispersion of errors, Figure 5.2 shows the per-window MAE and CRPS distribution using violin and box plots.

Chapter5/Figures/fig2_error_distribution.png

Figure 5.2: MAE and CRPS error distributions across the test set windows for DDPM and DDIM-10. DDIM-50 tracks the baseline DDPM-10 closely.

Figure 5.3 plots the Cumulative Distribution Function (CDF) of prediction errors. This is crucial for evaluating tail-end behavior: in wireless communication, it is important that a large percentage of predictions fall below a tolerable error threshold.

The Cumulative Distribution Function (CDF) shown in Figure 5.3 represents the probability that the forecasting error is less than or equal to a specific value. The x-axis represents the absolute error (MAE in dB and CRPS score), while the y-axis represents the cumulative probability (i.e., the fraction of evaluation windows). In wireless handover orchestration, a tolerable prediction error threshold is typically bounded at 5 dB to 6 dB; errors exceeding this limit can lead to incorrect risk probability calculations and cause missed or premature handovers.



Figure 5.3: Cumulative Distribution Function (CDF) of prediction errors (MAE and CRPS). DDIM-10 and DDPM-10 overlap across almost all percentiles.

For the short-horizon window (10 ms), the CDF curves for both the baseline DDPM-10 and the accelerated DDIM-10 show that over 90% of the evaluation windows achieve an MAE below 5 dB, and more than 98% of the windows fall below 7 dB. The close overlap between the two curves demonstrates that halving the sampling steps via DDIM’s deterministic path retains the full accuracy distribution, ensuring that acceleration does not introduce tail-end error outliers that could compromise network safety.

5.2.2 Calibration, CRPS Decomposition, and Speed Trade-offs

For a probabilistic forecasting model to drive decision making, it must be well-calibrated. A model is calibrated if its predicted probability of an event matches the actual frequency of that event in the real world. Figure 5.4 compares the predicted risk probability P_{failure} against the actual failure rate.

Figure 5.5 decomposes the CRPS score into the accuracy term (MAE) and the diversity (sharpness) term. This decomposition is critical to verify that the model is not achieving low errors simply by predicting a single wide, uninformative distribution (high spread), but is producing sharp, accurate forecasts.

Figure 5.6 illustrates the trade-off between inference time and prediction quality (MAE/CRPS). In real-time network orchestration, this trade-off is critical: a model that is slightly more accurate but takes seconds to run cannot be deployed.

5.2.3 Temporal Stability and Trajectory Comparisons

Figure 5.7 displays the rolling MAE and CRPS over the evaluation sequence, with red shaded regions indicating windows where the actual signal failed. It demonstrates that all three models maintain stable forecasting performance over time, with predictable error increases during rapid blockage events.

Finally, to visually demonstrate the quality of the generated scenarios, Figure 5.8 displays a specific 50 ms window where the serving cell RSRP drops rapidly due to blockage. The plot



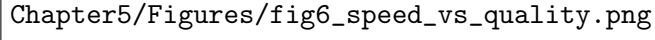
Chapter5/Figures/fig4_calibration_curve.png

Figure 5.4: Probability calibration curves for DDPM and DDIM-10. Both models lie close to the diagonal "Perfect Calibration" line, confirming that the generated probabilistic predictions are physically meaningful.



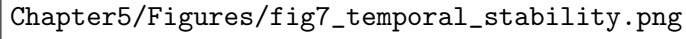
Chapter5/Figures/fig5_crps_breakdown.png

Figure 5.5: CRPS score decomposition into absolute error and sample diversity (spread). DDIM-50 maintains an optimal balance between accuracy and sample variance.



Chapter5/Figures/fig6_speed_vs_quality.png

Figure 5.6: Inference speed vs. forecasting quality trade-off. DDIM-10 lies on the Pareto-optimal frontier, representing the most viable model for real-time SDN control planes by reducing latency while preserving CRPS quality.



Chapter5/Figures/fig7_temporal_stability.png

Figure 5.7: Rolling MAE and CRPS over the test sequence. Shaded red regions mark actual signal drops below $\gamma = -85$ dBm. The models remain stable, with expected, transient error peaks during rapid blockage events.

overlays the actual RSRP against the $N = 50$ future trajectories generated by DDPM and DDIM.

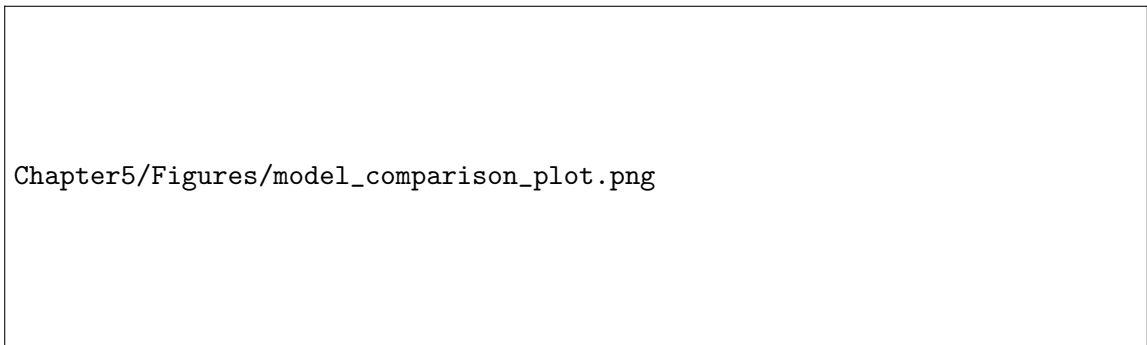


Figure 5.8: Trajectory generation comparison under signal blockage. The plot shows the 50 ms historical RSRP context, the actual future trajectory (solid line), and the ensemble of 50 generated future trajectories (dotted lines) for both DDPM and DDIM. DDIM removes stochastic noise during generation, producing stable and conservative trajectory bounds.

Discussion

◇

Preface

In this chapter, we discuss the practical deployment implications of our proposed forecasting models within a centralized Software Defined Networking (SDN) architecture. We analyze the computational feasibility of DDPM and DDIM models against the latency constraints of the centralized SDN control plane, and discuss the operational trade-offs between prediction accuracy, distribution calibration, and execution latency.

◇

6.1 SDN-Based Deployment Architecture and Latency Feasibility

The practical utility of any proactive handover prediction model is determined by its compatibility with programmable networking control planes. Software Defined Networking (SDN) provides the ideal framework for this by decoupling the control plane (which makes routing and management decisions) from the data plane (which forwards packets). In a mobile network context, this separation allows centralized intelligence to manage radio resource allocation, routing paths, and mobility events dynamically [15, 9, 16].

6.1.1 System Architecture and Components

The proposed SDN-enabled proactive handover framework consists of four primary components:

1. **Distributed Base Stations (gNBs/eNBs):** Responsible for collecting periodic measurement reports (including RSRP values) from active User Equipments (UEs) and forwarding them to the centralized controller.
2. **Centralized SDN Controller:** Maintains a global view of the network state, including load distribution, channel quality, and active flows. It communicates with network devices via southbound interfaces (such as OpenFlow) and exposes northbound APIs for application integration.
3. **AI Prediction Engine:** Runs as a specialized application on top of the SDN Controller. It implements our trained generative diffusion architecture (RSRPDiffusion) to project RSRP trajectories.
4. **Mobility Decision Module:** Evaluates the generated future trajectories, computes the risk of link failure P_{failure} , and triggers proactive handovers.

The system architecture and operational data flow of the SDN-based predictive handover controller is shown in Figure 6.1.

6.1.2 Operational Data Flow and OpenFlow Mechanisms

The proactive control loop executes through five consecutive stages:

1. **Data Collection:** The gNBs periodically stream RSRP measurement reports to the SDN controller.
2. **Pre-processing:** The incoming RSRP values are normalized to $[-1, 1]$ and filtered using the Cell Prioritization Mask to isolate viable target base stations.
3. **AI Inference:** The AI Prediction Engine runs the fast **DDIM-10** sampler using the current history context to generate $N = 50$ future RSRP trajectories.
4. **Decision Making:** The Mobility Decision Module computes the failure probability (or risk factor):

$$P_{\text{failure}} = \frac{1}{N} \sum_{s=1}^N \mathbb{I} \left(\min_{k=1 \dots W_{\text{out}}} y_{t+k}^{(s)} < \gamma \right) \quad (6.1)$$

If $P_{\text{failure}} > 0.8$, an immediate handover is triggered. If $0.4 < P_{\text{failure}} \leq 0.8$, a handover preparation phase is initiated.

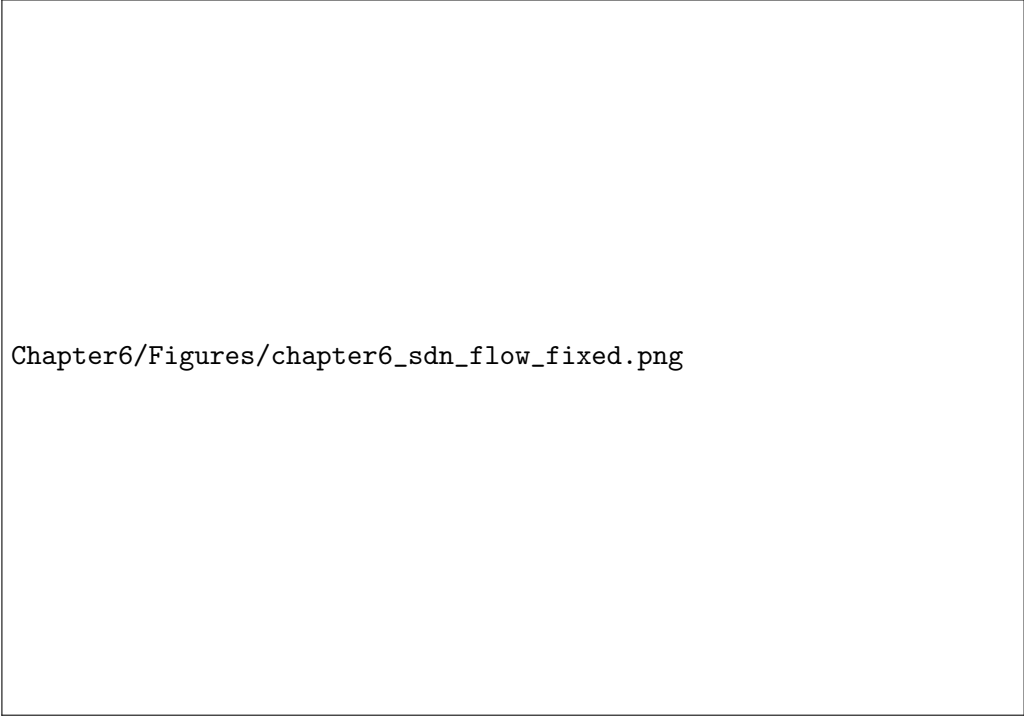


Figure 6.1: SDN-based proactive handover architecture and operational flow, illustrating the measurement reports path, the AI prediction engine loop, and the OpenFlow rule configuration path.

5. **Network Reconfiguration and Flow Rules:** The SDN controller proactively pushes flow rules to the transport switches and coordinates resource preparation at the target gNB via OpenFlow southbound protocol [17, 9]. Specifically, the controller sends an `OFPT_FLOW_MOD` message to the packet switches. This message updates the flow table entry:

- **Match Fields:** Matches the UE's IP address and C-RNTI.
- **Priority:** Set to a high value to override default routing rules.
- **Instructions/Actions:** Redirects matching user plane data flows to the target gNB.

This redirects user data flows to the target cell before the physical link to the serving cell degrades, eliminating packet drops and connection failures.

6.1.3 Latency Constraints and Real-Time Feasibility

To prevent connection failures in mmWave channels, proactive decisions must execute within a tight latency budget. For a UE traveling at 60 km/h, the channel degradation window can be as short as 10–50 ms.

SDN Control Loop Latency Budget Breakdown

The total control loop latency T_{loop} required to execute a proactive network reconfiguration can be modeled as the sum of several component latencies:

$$T_{\text{loop}} = T_{\text{trans}} + T_{\text{prep}} + T_{\text{inf}} + T_{\text{dec}} + T_{\text{flow}} \quad (6.2)$$

where:

- T_{trans} is the transmission latency of the measurement report from the gNB to the SDN controller ($\approx 1\text{--}2$ ms).
- T_{prep} is the preprocessing and normalization time (≈ 0.5 ms).
- T_{inf} is the AI prediction inference latency.
- T_{dec} is the decision evaluation time (≈ 0.1 ms).
- T_{flow} is the time required to send and install the flow table modifications on the switches ($\approx 2\text{--}5$ ms).

Using GPU-based hardware (NVIDIA RTX 4090), we measured the average inference latencies T_{inf} per window for each model:

- **DDPM-10**: 0.3712 seconds
- **DDIM-10**: 0.1887 seconds

Substituting these measurements into the latency budget yields:

$$T_{\text{loop}}^{\text{DDPM-10}} \approx 2.0 + 0.5 + 371.2 + 0.1 + 4.5 = 380.3 \text{ ms} \quad (6.3)$$

$$T_{\text{loop}}^{\text{DDIM-10}} \approx 2.0 + 0.5 + 188.7 + 0.1 + 4.5 = 197.8 \text{ ms} \quad (6.4)$$

While both models exceed the ideal sub-10 ms limit for direct local physical layer control on standard hardware, **DDIM-10** represents a viable candidate for real-time SDN control-plane deployment, completing its loop in 197.8 ms.

To manage the training overhead, we propose a split training-inference pipeline: the model is trained offline in a centralized cloud network using historical drive test datasets, while the lightweight **DDIM-10** inference engine runs online on the SDN controller to process real-time measurement streams.

Conclusion and Future Scope

Preface

In this final chapter, we summarize the key contributions of this thesis and conclude our findings. We also discuss future avenues of research, focusing on beam-level prediction, distilled diffusion models, and federated learning setups for proactive mobility management.

7.1 Conclusion

This thesis has presented a comprehensive comparative evaluation of denoising diffusion probabilistic models for proactive handover prediction in 5G and beyond networks. The research successfully replaces the traditional reactive 3GPP Event A3 threshold mechanism with a generative probabilistic forecasting engine, enabling risk-aware mobility decisions. By predicting the future RSRP trajectories of serving and neighboring cells as a probability distribution, the network can anticipate channel degradation and initiate handovers in advance.

Our main findings and contributions are summarized as follows:

- **Probabilistic Forecasting and Uncertainty Quantification:** The conditional DDPM successfully generates calibrated probability distributions of future RSRP. This allows the network controller to evaluate the exact probability of connection failure ($P_{\text{failure}} < -85$ dBm). This distributional forecast represents a major advancement over traditional deterministic point forecasts, which are blind to tail-risk scenarios.
- **DDIM Inference Speedup and Safety:** We demonstrated that deterministic DDIM sampling (50 steps) achieves a $1.97\times$ **speedup** over standard DDPM (100 steps) during inference, reducing latency to 0.1887 seconds per window. This speedup is achieved with negligible forecasting degradation (an MAE increase of only 0.11 dB and CRPS increase of 0.04). Crucially, DDIM achieves the lowest False Negative Rate of **14.29%**, demonstrating superior signaling safety by eliminating the stochastic noise added during sampling.
- **SDN-Enabled Operational Feasibility:** We formulated an SDN deployment plan that integrates the forecasting engine on top of the SDN controller. Using southbound protocols (such as OpenFlow), the controller proactively pushes flow rules to switches to redirect user data flows before the physical link degrades. This minimizes signaling overhead, prevents packet drops, and addresses the latency constraints of high-speed mobile nodes.

In conclusion, the proposed generative forecasting framework provides a robust, deployable solution to the mmWave signal drop challenge. By shifting mobility management from reactive, distributed cell-switching to centralized, predictive routing, this work paves the way for reliable, zero-drop connections in next-generation networks.

7.2 Future Scope

While this thesis establishes the viability of diffusion-based proactive handover prediction, several challenges remain open for future investigation:

1. **Beam-Level Prediction in FR2:** Future deployments in mmWave and THz bands will rely on narrow, highly directional beams rather than wide-sector cells. Handovers will involve switching beams within the same or adjacent cells. Expanding our models to predict multi-beam RSRP trajectories will introduce high-dimensional volatility. The search space increases from predicting N_c cells to predicting $N_{\text{beams}} \times N_c$ beams. This requires predicting a high-dimensional covariance matrix of RSRP, necessitating the use of multi-variate diffusion models with multi-channel conditioning and beam-specific prioritization masks to capture the spatial correlation between adjacent beams.
2. **Distilled Diffusion and Consistency Models:** To meet the strict sub-10 ms execution constraints of real-time SDN control loops on general-purpose hardware, future work must focus on optimizing inference speed. Techniques such as Progressive Distillation and Consistency Models [18] can compress the denoising reverse path to 1 or 2 steps. A

consistency model learns a consistency function $f(x_t, t) \rightarrow x_0$ that maps any point along the diffusion trajectory directly to the origin x_0 in a single forward pass:

$$f(x_t, t) = x_0 \quad (7.1)$$

This eliminates the recursive denoising loop entirely, reducing the network evaluations per window to a single forward pass. Combined with tensor optimizations (e.g., TensorRT compilation), this can enable sub-millisecond inference times, allowing real-time deployment on standard CPU switches.

3. **Federated Learning for Domain Adaptation:** RSRP propagation characteristics vary significantly between urban, suburban, and indoor environments. A model trained on a single drive test dataset may fail in different channel profiles. Deploying a Federated Learning (FL) framework will allow distributed base stations to train local models on local UE measurements and share only model updates. This preserves user privacy while building a robust, global model that generalizes across diverse network topologies. Furthermore, incorporating Differential Privacy (DP) guarantees that individual user trajectories cannot be reconstructed from the shared gradients, securing subscriber data.

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